Yunhan (Jack) Jia

Founder, Trenz · AI agents for social commerce · Previously ML security research at ByteDance and Baidu

Sunnyvale, CA
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EXPERIENCE

Founder · Trenz, Sunnyvale, CA — 06/2024 – present

- Building an AI-native social commerce platform: a coordinated team of AI agents that handles market research, content creation and multi-platform publishing for TikTok Shop and social sellers.

Research Scientist · AI Lab, ByteDance, US — 08/2019 – 2024

Senior Security Scientist · X-Lab, Baidu USA — 04/2018 – 08/2019

- Led AI security research: physical adversarial attacks on perception, DNN robustness benchmarking (Perceptron Benchmark), structure-aware fuzzing of Baidu Apollo.

Threat Research Engineer Intern · Palo Alto Networks — 05/2017 – 08/2017

Research Intern · QoE Lab, T-Mobile US — Summers 2014, 2015

Intern · CSS Lab, Microsoft APGC — 08/2012 – 09/2012

EDUCATION

Ph.D., Computer Science and Engineering · University of Michigan, Ann Arbor — 09/2013 – 04/2018

Advisor: Prof. Z. Morley Mao. Thesis: Program Analysis based Approaches to Ensure Security and Safety of Emerging Software Platforms

B.E., Software Engineering · Shanghai Jiao Tong University — 09/2009 – 06/2013

Advisor: Prof. Haibo Chen

PUBLICATIONS

Dirty Road Can Attack: Security of Deep Learning based Automated Lane Centering under Physical-World Attack

Takami Sato, Junjie Shen, Ningfei Wang, **Yunhan Jia**, Xue Lin, Qi Alfred Chen
`USENIX Security` 2021

Fooling Detection Alone is Not Enough: Adversarial Attack against Multiple Object Tracking

Yunhan Jia, Yantao Lu, Junjie Shen, Qi Alfred Chen, Hao Chen, Zhenyu Zhong, Tao Wei
`ICLR` 2020 · earlier version at CVPR AdvMLCV Workshop 2019 (oral)

Enhancing Cross-Task Black-Box Transferability of Adversarial Examples with Dispersion Reduction

Yantao Lu, **Yunhan Jia**, Jianyu Wang, Bai Li, Weiheng Chai, Lawrence Carin, Senem Velipasalar
`CVPR` 2020 · earlier version at CVPR AdvMLCV Workshop 2019

Explainable Machine Learning in Deployment

Umang Bhatt, Alice Xiang, Shubham Sharma, Adrian Weller, Ankur Taly, **Yunhan Jia**, Joydeep Ghosh, Ruchir Puri, José M. F. Moura, Peter Eckersley
`FAT*` ACM Conference on Fairness, Accountability, and Transparency, 2020

Towards Practical Lottery Ticket Hypothesis for Adversarial Training

Bai Li, Shiqi Wang, **Yunhan Jia**, Yantao Lu, Zhenyu Zhong, Lawrence Carin, Suman Jana
`arXiv` 2020

Attacking Multiple Object Tracking using Adversarial Examples

Yunhan Jia, Yantao Lu, Junjie Shen, Qi Alfred Chen, Zhenyu Zhong, Tao Wei
`ICML SPML` Workshop, 2019

The Cost of Learning from the Best: How Prior Knowledge Weakens the Security of Deep Neural Networks

Yunhan Jia, Zhenyu Zhong, Yulong Zhang, Qian Feng, Tao Wei `Black Hat Asia` 2019

Perception Deception: Physical Adversarial Attack Challenges and Tactics for DNN-based Object Detection
Yunhan Jia, Zhenyu Zhong, Weilin Xu, Tao Wei `Black Hat Europe` 2018

Dissecting HAS VOD Services for Cellular: Performance, Root Causes and Best Practices
Shichang Xu, **Yunhan Jia**, Z. Morley Mao, Subhabrata Sen `IMC` 2017

TrafficNet: An Open Naturalistic Driving Scenario Library
Ding Zhao, Yaohui Guo, **Yunhan Jia** `ITSC` 2017

Towards Secure and Safe Appified Automated Vehicles
Yunhan Jia, Ding Zhao, Qi Alfred Chen, Z. Morley Mao `IEEE IV` 2017

ContexIoT: Towards Providing Contextual Integrity to Appified IoT Platforms
Yunhan Jia, Qi Alfred Chen, Shiqi Wang, Amir Rahmati, Earlence Fernandes, Z. Morley Mao, Atul Prakash `NDSS` 2017

Open Doors for Bob and Mallory: Open Port Usage in Android Apps and Security Implications
Yunhan Jia, Qi Alfred Chen, Yikai Lin, Chao Kong, Z. Morley Mao `IEEE EuroS&P` 2017

The Misuse of Android Unix Domain Sockets and Security Implications
Yuru Shao, Jason Ott, **Yunhan Jia**, Zhiyun Qian, Z. Morley Mao `ACM CCS` 2016

Static Detection of Packet Injection Vulnerabilities: A Case for Identifying Attacker-controlled Implicit Information Leaks
Qi Alfred Chen, Zhiyun Qian, **Yunhan Jia**, Yuru Shao, Z. Morley Mao `ACM CCS` 2015

Performance Characterization and Call Reliability Problem Diagnosis for Voice over LTE
Yunhan Jia, Qi Alfred Chen, Z. Morley Mao, Jie Hui, Kranthi Sontineni, Alex Yoon, Samson Kwong, Kevin Lau `ACM MobiCom` 2015

TALKS

- Attacking Multiple Object Tracking using Adversarial Examples — CVPR Workshop — 06/2019
- Threat to Real-world Deep Learning Systems: Practical Attacks and Security Measures — UC Irvine — 04/2019
- The Cost of Learning from the Best — Black Hat Asia — 03/2019
- Perception Deception — Black Hat Europe — 12/2018
- From Memory Safety to AI Security — USENIX Security — 08/2018
- Lessons Learnt from Securing Modern IoT and Autonomous Vehicle Platforms — Palo Alto Networks — 07/2017
- SmartPhone, SmartHome, and SmartCar: Lessons from Securing Appified Platforms — SJTU — 01/2017
- Conference talks at IEEE IV, IEEE EuroS&P, NDSS (2017) and ACM MobiCom (2015)

PATENTS

- Monitoring Wireless Data Consumption — US 2017/0005989
- Cross-Layer Link Failure Alerts — US 2016/0065433
- Pathway-based Data Interruption Detection — US 2016/0277952
- Indoor/Outdoor Coverage Detection Based on Sensor Inputs — US 2017/0366943
- Machine Learning-based Troubleshooting of VoLTE Calls — US 2018/0192303

HONORS & SERVICE

- Internet of Things Technology Research Award, Google — 2016
- CSE Fellowship, University of Michigan — 2014
- Excellent Participation in Research Program, Shanghai Jiao Tong University — 2011
- Program Committee: AutoSec, CVPR AdvMLCV, ICML SPML — 2019